

Class 12 Chemistry NCERT Solutions

Chapter 1: Solutions

Problem 1.1

Calculate the mass percentage of benzene (C_6H_6) and carbon tetrachloride (CCl_4) if 22 g of benzene is dissolved in 122 g of carbon tetrachloride.

Solution: Mass of solution = Mass of C_6H_6 + Mass of CCl_4 Mass of solution = 22 g + 122 g = 144 g

Mass percentage of C_6H_6 = $\frac{\text{Mass of } C_6H_6}{\text{Total mass of solution}} \times 100$ Mass percentage of C_6H_6 = $\frac{22}{144} \times 100 = 15.28\%$

Mass percentage of CCl_4 = $\frac{\text{Mass of } CCl_4}{\text{Total mass of solution}} \times 100$ Mass percentage of CCl_4 = $\frac{122}{144} \times 100 = 84.72\%$

Problem 1.2

State Raoult's Law. **Answer:** Raoult's law states that for a solution of volatile liquids, the partial vapour pressure of each component of the solution is directly proportional to its mole fraction present in solution. For a binary solution of components A and B:

$$P_A = P_A^0 \chi_A$$

$$P_B = P_B^0 \chi_B$$

Chapter 2: Electrochemistry

Problem 2.1

Represent the cell in which the following reaction takes place: $Mg(s) + 2Ag^+(0.0001M) \rightarrow Mg^{2+}(0.130M) + 2Ag(s)$ Calculate its E_{cell} if $E_{cell}^0 = 3.17$ V.

Solution: The cell representation is: $Mg(s)|Mg^{2+}(0.130M)||Ag^+(0.0001M)|Ag(s)$ Using Nernst Equation:

$$E_{cell} = E_{cell}^0 - \frac{0.0591}{n} \log \frac{[Mg^{2+}]}{[Ag^+]^2}$$

Here $n = 2$.

$$E_{cell} = 3.17 - \frac{0.0591}{2} \log \frac{0.130}{(0.0001)^2}$$

$$E_{cell} = 3.17 - 0.02955 \log(1.3 \times 10^7)$$

$$E_{cell} = 3.17 - 0.02955(7.1139)$$

$$E_{cell} = 3.17 - 0.21 = 2.96 \text{ V}$$

Chapter 3: Chemical Kinetics

Problem 3.1

The rate constant for a first order reaction is 60 s^{-1} . How much time will it take to reduce the initial concentration of the reactant to its $1/16^{th}$ value?

Solution: For a first order reaction:

$$t = \frac{2.303}{k} \log \frac{[R]_0}{[R]}$$

Given $k = 60 \text{ s}^{-1}$ and $[R] = \frac{[R]_0}{16}$.

$$t = \frac{2.303}{60} \log \frac{[R]_0}{[R]_0/16}$$

$$t = \frac{2.303}{60} \log 16$$

$$t = \frac{2.303}{60} \log 2^4 = \frac{2.303 \times 4 \times 0.3010}{60}$$

$$t = \frac{2.773}{60} = 0.0462 \text{ s} = 4.62 \times 10^{-2} \text{ s}$$

Chapter 4: d- and f- Block Elements

Problem 4.1

Why do transition elements exhibit variable oxidation states? **Answer:** Transition elements exhibit variable oxidation states because the energy difference between $(n-1)d$ and ns orbitals is very small. As a result, electrons from both energy levels can participate in bond formation.

Problem 4.2

Calculate the 'spin-only' magnetic moment of $M^{2+}(aq)$ ion ($Z = 27$). **Answer:** $Z = 27$ is Cobalt (Co). Electronic configuration of $Co = [Ar]3d^7 4s^2$. Electronic configuration of $Co^{2+} = [Ar]3d^7$. Number of unpaired electrons (n) in $3d^7$ ($t_{2g}^5 e_g^2$): d^7 has 3 unpaired electrons. Magnetic moment $\mu = \sqrt{n(n+2)} \text{ BM}$ $\mu = \sqrt{3(3+2)} = \sqrt{15} = 3.87 \text{ BM}$.